

Unit 10, Lesson 1: Sectors



Benchmark

MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.



Learning Targets

- ☐ I can define a sector of a circle.
- ☐ I can explain how the area of a sector is defined as a fractional part of the area of a circle.



10.1.1 Warm-Up: Pac-Man

The classic and popular Pac-Man video game was released in Japan in May of 1980 and was in the United States by October of that same year. The yellow, pie-shaped character travels around a maze trying to eat dots while avoiding the ghosts hunting him.

Watch the video of Pac-Man going around the maze and eating dots.



1. What do you notice about Pac-Man's body as he opens and closes his mouth to eat the dots?
2. What do you wonder as you watch Pac-Man move around the maze?



10.1.2 Exploration: Fractions of a Circle

Work with your partner to complete the following.

1. Pizza sizes are given in terms of the pizza's diameter. The standard size for most medium circular pizzas is 14 inches (in.) and it is usually cut into 10 triangular slices.



- a. What is the approximate area of a 14-inch pizza? Use $\frac{22}{7}$ for π .
- b. If the slices are cut precisely and equally, what fraction and percentage of the pizza is 5 slices?

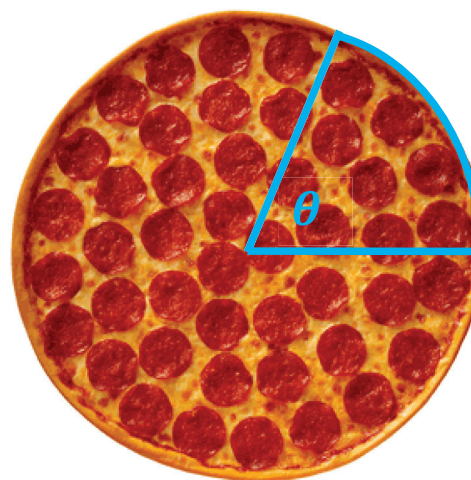
Fraction	Percentage

- c. What is the area of 5 slices?
- d. What is the area of 1 slice?

2. The point where a pizza is divided to create precisely equal slices is a central angle.



A **central angle** is an angle with its vertex at the center of a circle with radii as its side.



The central angle is a fraction of the total 360° in a circle.

- a. Complete the table. The first row of the table has been filled in.

	Fraction	Central Angle in Degrees ($^\circ$)
Half of a Circle	$\frac{1}{2}$	$\left(\frac{1}{2}\right) \cdot (360^\circ) = 180^\circ$
Quarter of a Circle		
Third of a Circle		
Tenth of a Circle		

b. Complete the statement.

One slice, or fraction of the circle, is bounded by two
☐ diameters
☐ radii and an arc. Since each slice is the same
 size, the central angle for each fractional piece of the
 circular pizza will ☐ vary.
☐ be equal.

c. The pizza slices are an example of **sectors**. With your partner, write a definition of the term **sector**.

d. Ask two other partner pairs to share their definition of a sector. Record a summary of their definition. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was correct.

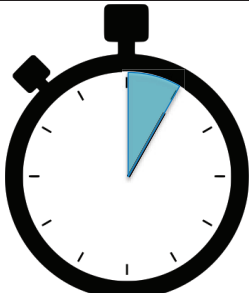
My Summary of Their Definition of a Sector	Initials



10.1.3 Guided Practice: Finding the Area of a Sector

The images below show a stopwatch with different time increments marked. The blue shading on the stopwatches each forms a sector.

1. Complete the table.

	Sector Fraction of the Circle	Central Angle Measure of the Sector	Ratio of the Central Angle to the Whole Circle
			
			
			

- Summarize what you notice about the relationship between the values in each row.

- Discuss with your partner how ratios could be used to find the area of the sectors, including any additional information that might be needed. Summarize your discussion.

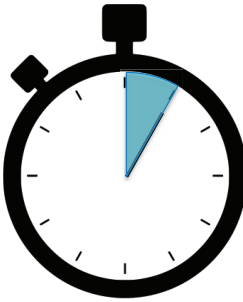
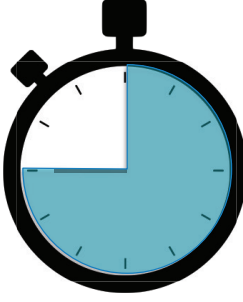
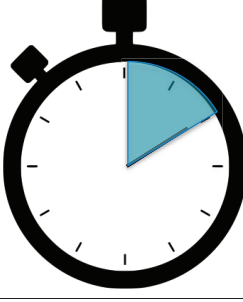
Given the radius of the stopwatch is 2 in., the area of the stopwatch is 4π in.².

$$A = \pi r^2$$

$$A = \pi(2)^2$$

$$A = 4\pi \text{ in.}^2$$

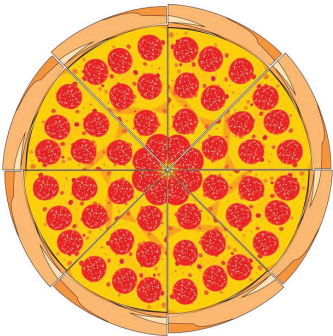
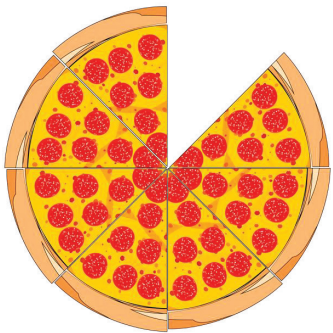
- Use the relationship between each sector and the whole stopwatch explored in problem 1 to find the area of each sector. Show your work.

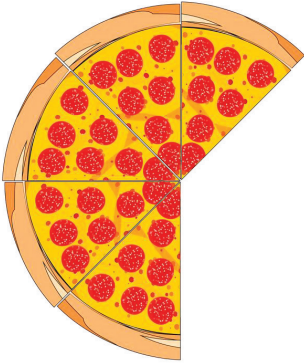
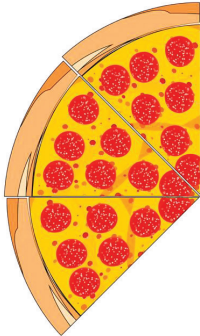
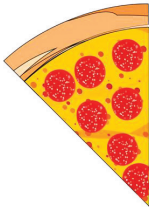
	Area of the Blue-Shaded Sector
	
	
	



10.1.4 Your Turn!

1. Kenya orders a large pizza that has a 16 in. diameter. Complete the table on the number of slices she has.

	Fraction of the Circle	Area of the Sector	Central Angle
Eight Slices 			
Seven Slices 			

	Fraction of the Circle	Area of the Sector	Central Angle
Five Slices 			
Three Slices 			
One Slice 			



10.1.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about sectors.

Unit 10, Lesson 2: Arc Length



Benchmark

MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.



Learning Targets

- ☐ I can define arc length.
- ☐ I can explain how the length of an arc is proportional to the ratio of the central angle to the whole circle, measured in degrees.